# Shubhadeep Mandal

81602 63797 shubhadeepmandal11@gmail.com shubhadeep-mandal Seronic2001

## Education

**International Institute of Information Technology, Hyderabad** July 2025 – Present
*Telangana, India*
*Master of Technology in Computer Science and Engineering* *CGPA: 9.12*

**Indian Institute of Information Technology, Ranchi** Aug 2020 – Jun 2024
*Ranchi, India*
*Bachelor of Technology (Hons.) in Computer Science and Engineering* *CGPA: 9.05*

## Projects

**RookDB: Vectorized Relational Storage Engine** | *Rust, Database Internals, B+Tree, CBO, Volcano*

- Co-architected a multi-crate SQL-99 RDBMS in Rust verified by **700+ value-level tests**, engineering slotted-page storage, CLOCK buffer pools, B+Tree indexing, and Volcano query execution.
- Engineered vectorized Volcano query operators, a Cost-Based Optimizer with HyperLogLog statistics, and Grace Hash Join, executing **2M-match joins in 5.85s** and streaming scans at **16.9M rows/s**.
- Designed an executor-tier cache and LRU query plan cache, eliminating 8 per-row I/O bottlenecks and driving a **100–400x DML throughput turnaround** (38,986 inserts/sec vs. 94 baseline) at 10M-row scales.

**Peer-to-Peer Distributed File Sharing System** | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Engineered a BitTorrent-style hybrid P2P system in C++, achieving an **8.5x transfer speedup on 1GB files** (193s to 22.8s) across 5 concurrent peers by saturating network and disk I/O.
- Implemented out-of-order parallel chunk writing directly to designated file offsets and dynamic bitfield generation for real-time partial seeding during active downloads.
- Architected Primary-Backup tracker failover with 2s bounded failover, full state synchronization, and verified resilience via **48 unit tests** under AddressSanitizer/UBSan.

**Distributed Analytics: MapReduce & gRPC Streaming** | *Hadoop Streaming, C++, gRPC, SLURM, YARN*

- Architected a real-time stream engine over bidirectional gRPC streams sustaining **123k records/s**, answering concurrent analytical queries in **2.1–4.5 ms** mid-ingestion via ack-delta state mirroring and bounded backpressure.
- Engineered C++ MapReduce streaming mappers with in-memory block flushes, proving commutative/associative partial merges to collapse batch processing to a single stage where **zero raw records cross the network shuffle**.
- Implemented distributed matrix multiplication on Hadoop Streaming with distributed-cache broadcast replication and combiner pre-aggregation, handling uneven row-partition splits and sparse matrix inputs across YARN clusters.

**FOOL: Custom POSIX-Compliant Interactive Shell** | *C++, POSIX APIs, Job Control, Termios, Linux*

- Built an interactive POSIX shell in C++ supporting multi-stage pipelines, I/O redirections, and wildcard expansion, validated via an automated **98-check test suite**.
- Eliminated signal race conditions and lost exit statuses using the self-pipe trick for async-safe `SIGCHLD` handling and centralizing process reaping on the main thread.
- Accelerated startup by **52x** (5.2s to 0.1s on WSL) via directory entry caching; built zero-dependency raw terminal editor using POSIX `termios` with tab autocompletion and persistent history.

## Technical Skills

**Languages**: C/C++, Rust, Python, TypeScript, JavaScript, SQL, Bash.

**Systems & OS**: POSIX APIs, Sockets, Linux/Unix, Multithreading, Signals, Termios, OpenSSL, ASan/UBSan.

**Distributed Systems & Storage**: Hadoop Streaming, HDFS/YARN, gRPC, Protobuf, SLURM, Storage Engines, Volcano Model.

**AI, Web & Tools**: PyTorch, Transformers, Scikit-learn, Next.js, React, Convex, PostgreSQL, Git, Docker.

## Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

## Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.